Enteral and Parenteral Nutrition

Nutrition assessment • Requirements • Enteral nutrition • Parenteral nutrition • Refeeding syndrome — all key points for the exam, with drug tables

1. Nutrition Assessment

Malnutrition

- **ASPEN/AND criteria:** ≥ 2 of: insufficient **energy intake**, **weight loss**, loss of **muscle mass**, loss of **subcutaneous fat**, **fluid accumulation**, ↓ **grip strength** • GLIM criteria also used.
- **Significant unintentional weight loss: > 5% in 1 month**, **> 7.5% in 3 months**, **> 10% in 6 months**, > 20% in 1 year.
- **Screening tools:** NRS-2002, MUST, **NUTRIC** score (ICU) → identify **high nutrition risk**.
- **Serum proteins** (**albumin** t½ ~20 days, **prealbumin** 2–3 days, transferrin 8–10 days) are **negative acute-phase reactants** → fall with inflammation, infection, fluid shifts → **NOT reliable markers of nutrition status** (ASPEN) • prealbumin can track response if CRP is normal.

Body weight

Weight	Formula / use
Ideal body weight (IBW)	Men: **50 kg + 2.3 kg per inch over 5 feet** • Women: **45.5 kg + 2.3 kg per inch over 5 feet**
Adjusted body weight (AdjBW)	IBW + **0.25** (or 0.4) × (actual − IBW) • used when actual weight > ~120–130% of IBW
Actual body weight	Use if less than IBW
BMI	kg/m² • < 18.5 underweight • 25–29.9 overweight • ≥ 30 obese

2. Nutrition Requirements

Nutrient	Requirement
Energy (general)	**25–30 kcal/kg/day** • predictive equations (Harris-Benedict, Mifflin-St Jeor, Penn State in ventilated) • **indirect calorimetry** = gold standard
Energy (critically ill)	**12–25 kcal/kg/day** during the first 7–10 days (ASPEN/SCCM 2022), then advance • avoid overfeeding
Energy (obese ICU)	**Hypocaloric, high-protein**: BMI 30–50 → **11–14 kcal/kg actual weight** • BMI > 50 → **22–25 kcal/kg IBW**
Protein (healthy)	**0.8 g/kg/day**
Protein (hospitalized, stress, ICU)	**1.2–2.0 g/kg/day** • burns, trauma up to 2–2.5 g/kg/day
Protein (obese ICU)	**2 g/kg IBW** (BMI 30–40) • **up to 2.5 g/kg IBW** (BMI ≥ 40)
Protein (kidney)	CKD not on dialysis: 0.6–0.8 g/kg • **hemodialysis 1.0–1.2** (up to 1.5) • **CRRT up to 2.5 g/kg** • **AKI: do not restrict protein** to delay dialysis
Protein (liver)	**1.2–1.5 g/kg** • **no protein restriction in hepatic encephalopathy**
Fluid	**30–35 mL/kg/day** (~1 mL/kcal) • more with fever, losses; less in HF/renal failure

Caloric values (high-yield)

Substrate	kcal
Dextrose (IV, monohydrate)	**3.4 kcal/g**
Carbohydrate (oral/enteral)	4 kcal/g
Protein / amino acids	**4 kcal/g**
Fat (dietary)	9 kcal/g
Lipid injectable emulsion	**10% = 1.1 kcal/mL • 20% = 2 kcal/mL • 30% = 3 kcal/mL** (includes glycerol and phospholipid)
Propofol	**1.1 kcal/mL** (10% lipid) → count in daily calories • **clevidipine** 2 kcal/mL (20% lipid)
Dextrose-containing IV fluids	D5W = 170 kcal/L (count IV fluids and drug diluents)

Nitrogen balance

- **Nitrogen balance** = protein intake (g) ÷ **6.25** − (**24-h urine urea nitrogen** + **4**) • goal **+2 to +4 g/day** (anabolic) • 1 g nitrogen = 6.25 g protein.

3. Choosing the Route: Enteral vs Parenteral

> **"If the gut works, use it."** → **Enteral nutrition (EN) is preferred** over parenteral nutrition (PN): maintains gut integrity and immune function, ↓ infections, fewer complications, cheaper.

- Start EN within **24–48 hours** in critically ill, hemodynamically stable patients who cannot eat.
- **Contraindications to EN: bowel obstruction**, severe **ileus**, **high-output fistula**, severe **GI bleeding**, intractable vomiting/diarrhea, **mesenteric ischemia**, **hemodynamic instability** (escalating vasopressors), severe short bowel syndrome, peritonitis.
- **Oral nutrition supplements** first if the patient can eat but intake is inadequate.

4. Enteral Nutrition (EN)

Access

Duration	Gastric	Post-pyloric (small bowel)
Short term (< 4–6 weeks)	**Nasogastric (NG)** / orogastric	**Nasoduodenal / nasojejunal (NJ)**
Long term (> 4–6 weeks)	**Gastrostomy (PEG**, surgical G-tube)	**Jejunostomy** (PEJ, surgical J-tube), PEG-J

- **Gastric feeding** = more physiologic, allows **bolus** feeding, gastric acid kills bacteria • **post-pyloric** = for **high aspiration risk**, gastroparesis, gastric outlet problems, intolerance in pancreatitis → requires **continuous** feeding (no bolus into the jejunum → dumping).
- **Confirm tube position by X-ray** before the first use of a blindly placed tube.

Administration methods

Method	Description
Continuous	Pump over 24 h • ICU, post-pyloric, intolerance • start **10–40 mL/h**, ↑ by **10–20 mL/h every 4–8 h** to goal
Intermittent	Gravity or pump over 20–60 min several times daily
Bolus	240–480 mL over 5–10 min by syringe 4–6 times daily • **gastric only** • home/ambulatory
Cyclic	Over 8–16 h (often overnight) • allows daytime oral intake/activity

Enteral formulas

Type	Features / use
Standard polymeric	Intact protein, **1–1.2 kcal/mL** • most patients
Calorically dense	**1.5–2 kcal/mL** • **fluid restriction** (heart failure, renal failure, SIADH)
High-protein	Critically ill, wounds, obese ICU
Elemental / semi-elemental (peptide-based)	Hydrolyzed protein, low fat/MCT • **malabsorption**, short bowel, pancreatitis, intolerance
Renal	Calorie-dense, **low K, phosphate, Mg**, low volume (e.g., Nepro, Suplena)
Diabetes	Lower carbohydrate, more fat/fiber
Hepatic	Branched-chain amino acid enriched (limited evidence)
Pulmonary	High fat/low carb (to ↓ CO_2) — **not recommended** (avoid overfeeding instead)
Immune-modulating	Arginine, glutamine, omega-3, nucleotides • perioperative/trauma • **avoid arginine in sepsis**
Fiber-containing	Helps diarrhea/constipation • avoid in bowel ischemia or severe dysmotility
Modular	Protein, carbohydrate, or fat added to a base formula

- **Free water:** formulas are ~**80–85% water** (1 kcal/mL) to ~**70%** (2 kcal/mL) → add **water flushes** to meet fluid needs (watch for **hypernatremia** with high-protein, concentrated formulas).

Complications of EN

Type	Problems and management
Mechanical	Malposition (X-ray), **tube clogging** → flush with **30 mL warm water** before/after feeds and medications and **every 4 h** during continuous feeds • unclog with warm water, then **pancreatic enzyme + sodium bicarbonate** (not cola or cranberry juice) • dislodgement, sinusitis, skin irritation
Aspiration (most serious)	**Elevate head of bed 30–45°** • post-pyloric feeding • continuous instead of bolus • **prokinetics** (**metoclopramide** 10 mg q6h; **erythromycin** 250 mg IV q6–8h — tachyphylaxis after ~72 h) • ASPEN: **do not use gastric residual volume routinely**; do not hold feeds for GRV < 500 mL without other signs of intolerance
Diarrhea (most common)	**Check medications first**: **sorbitol-containing liquids** (acetaminophen, guaifenesin, theophylline elixirs), **magnesium**, antibiotics, **C. difficile**, laxatives, prokinetics • then formula: osmolality, rate, fat malabsorption → add fiber, peptide formula, slow rate • antidiarrheals only after excluding infection
Other GI	Nausea, vomiting, distension, constipation (fluid, fiber)
Metabolic	**Hyperglycemia**, electrolyte disorders, **refeeding syndrome**, dehydration/**hypernatremia** (too little free water), fluid overload

Medication administration through feeding tubes (high-yield)

- **Do NOT crush: enteric-coated** (EC), **extended-release** (ER, SR, XL, CR, LA), **sublingual/buccal**, **hazardous drugs** (chemotherapy, hormones — exposure risk), some capsules with special beads.
- Prefer **liquid** formulations (watch **sorbitol** and **high osmolality** → dilute) or **crush immediate-release tablets** finely and mix with 15–30 mL water.
- Give **each drug separately** • **flush 15–30 mL water before, between, and after** each drug • **never add drugs to the formula bag**.
- **Jejunal tubes:** avoid hyperosmolar medications (diarrhea, dumping) and drugs needing gastric acid or gastric absorption.
- Drugs that **clog tubes**: sucralfate, cholestyramine, psyllium, oil-based suspensions.

Drug	Interaction with enteral feeds	Management
Phenytoin	Binds formula/tube → ↓ levels (up to 70%)	**Hold feeds 1–2 h before and after** • monitor levels • adjust dose when feeds start/stop
Warfarin	Binds protein + vitamin K in formula → **warfarin resistance**	Hold feeds ~1 h before/after • **monitor INR** closely when feeds change

Drug	Interaction with enteral feeds	Management
Fluoroquinolones (cipro, levo, moxi)	**Chelation** with Ca, Mg, Fe in formula → ↓ absorption	Hold feeds 1–2 h before and after • crush tablets — **do not give oral ciprofloxacin suspension** via tube (oily, clogs)
Tetracyclines	Chelation	Separate from feeds
Levothyroxine	↓ absorption	Hold feeds ~1 h before and 30–60 min after • monitor TSH
Carbamazepine suspension	Adsorbs to PVC tubing	Dilute before giving
PPIs	EC granules	Use **suspension packets** (omeprazole, esomeprazole, pantoprazole granules in apple juice) or **lansoprazole ODT** dispersed in water

5. Parenteral Nutrition (PN)

Indications and timing

- **GI tract not functional** or cannot meet needs: **severe short bowel syndrome**, **bowel obstruction**, prolonged **ileus**, **high-output fistula**, severe malabsorption, severe mucositis, hyperemesis, **failure of EN** • usually expected need **≥ 7 days**.
- **Critically ill (ASPEN/SCCM): low nutrition risk**, well nourished → **wait 7 days** before starting PN if EN is not possible • **high nutrition risk or severely malnourished** → start PN **as soon as possible** • **supplemental PN** after 7–10 days if EN provides < 60% of goal.

Central vs peripheral PN

Feature	Central PN (CPN)	Peripheral PN (PPN)
Access	Central venous catheter (PICC, tunneled catheter, port) — tip in SVC	Peripheral vein
Osmolarity	High (up to ~1,800–2,000+ mOsm/L)	**≤ 900 mOsm/L** (phlebitis)
Dextrose concentration	Up to 70% stock (final ~15–25%)	Low (~5–10% final)
Can meet full needs?	**Yes** (concentrated)	Often **no** — large volumes needed; lipids help (isotonic)
Use	Most PN patients, long term, fluid restriction	**Short term (≤ 1–2 weeks)**, mild needs, no fluid restriction

- **Estimating PN osmolarity** (mOsm/L): **dextrose ~5 mOsm/g**, **amino acids ~10 mOsm/g**, **electrolytes ~2 mOsm/mEq** (lipids add little).

PN formulation types

- **2-in-1** (dextrose + amino acids; lipid infused separately) → **0.22-micron** filter • can see precipitates.
- **3-in-1 / total nutrient admixture (TNA)** (lipids mixed in) → **1.2-micron** filter • less stable, **cannot see precipitates** (opaque) • watch for "**cracked**" emulsion (oil separation) → do not use.
- **Hang time:** lipid alone **12 h** • TNA and 2-in-1 **24 h**.
- **Commercial premixed** (e.g., Clinimix, Kabiven/Perikabiven multichamber bags) vs **compounded** (individualized).

Macronutrients

Component	Key points
Dextrose (3.4 kcal/g)	Main energy source • start ~**150–200 g on day 1** (or ~50% of goal), advance as glucose allows • **max glucose infusion rate 4–5 mg/kg/min** (≤ 4 in critically ill) • excess → **hyperglycemia**, **hepatic steatosis**, ↑ **CO2 production** (difficult ventilator weaning) • glucose target **140–180 mg/dL** → add **regular insulin to the PN bag** (e.g., start ~0.1 unit per g dextrose) ± sliding scale
Amino acids (4 kcal/g)	Standard (10–15% stock) • specialty: **renal** (essential AAs), **hepatic** (BCAA), **pediatric** (TrophAmine — contains taurine, tyrosine; add cysteine)
Lipid injectable emulsions (ILE)	Provide calories and **essential fatty acids** • usually **20–30% of total calories** • **≤ 1–1.5 g/kg/day** (max 2.5 g/kg/day adults; soybean ≤ 1 g/kg/day in ICU/liver disease) • hold or reduce if **TG > 400 mg/dL**

Lipid emulsion products

Product	Oil source	Notes
Intralipid, Nutrilipid	**Soybean** (100%)	High **omega-6** (pro-inflammatory) • **phytosterols** → linked to **PN-associated liver disease** (especially children)
SMOFlipid	**S**oybean, **M**CT, **O**live, **F**ish	Balanced • less liver injury
Clinolipid	Olive (80%) + soybean (20%)	Less omega-6
Omegaven	**Fish oil** (100%)	Pediatric **PN-associated cholestasis** • dose 1 g/kg/day • low essential fatty acids alone

- **Contraindications:** severe **egg** (egg phospholipid emulsifier), **soy**, or **fish** allergy (product-dependent) • severe hypertriglyceridemia (TG > 1,000) • hyperlipidemia-induced pancreatitis.
- **Essential fatty acid deficiency (EFAD)** after ~1–3 weeks lipid-free → **dry, scaly rash**, alopecia, poor wound healing, thrombocytopenia • **triene:tetraene ratio > 0.2** • prevent with ~**2–4% of calories as linoleic acid** (e.g., soybean 20% 250 mL 2–3 times/week).
- **Infusion rate** ≤ 0.11 g/kg/h • ADRs: hypertriglyceridemia, fat overload syndrome, infection risk, liver disease • **propofol** counts as lipid.

Micronutrients

Component	Daily adult amount / key points
Sodium	**1–2 mEq/kg**
Potassium	**1–2 mEq/kg**
Calcium	**10–15 mEq** (as **calcium gluconate** — not chloride)
Magnesium	**8–20 mEq**
Phosphate	**20–40 mmol**
Acetate / chloride	As needed for acid–base • **acetate** (→ bicarbonate) for **metabolic acidosis** • **chloride** for alkalosis • **NEVER add bicarbonate** (precipitates with calcium → calcium carbonate; CO2)
Multivitamins	**Daily** (13 vitamins) • adult product contains **vitamin K 150 mcg** → affects **warfarin** • during shortages prioritize; give **thiamine**
Trace elements	Daily: **zinc**, **copper**, **manganese**, **selenium**, (chromium) • ↑ **zinc** with diarrhea, fistulas, burns • ↓ **copper and manganese** in **cholestasis** (biliary excretion; manganese neurotoxicity) • ↓ **chromium, selenium** in renal failure • ↑ selenium in burns/CRRT • **iron not routinely added** (incompatible with TNA)
Aluminum	Contaminant (calcium gluconate glass vials, phosphate salts) • limit **< 5 mcg/kg/day** in neonates and renal impairment

- **Trace element deficiencies: zinc** → rash (acrodermatitis-like), poor wound healing, diarrhea, alopecia • **copper** → anemia, neutropenia, myeloneuropathy • **selenium** → cardiomyopathy, muscle weakness • **chromium** → hyperglycemia, neuropathy • **thiamine** → Wernicke's, lactic acidosis, wet beriberi (heart failure).

Calcium–phosphate compatibility (high-yield)

- **Calcium-phosphate precipitation** is the most important PN compatibility problem → can cause **fatal pulmonary emboli** (FDA safety alert).
- **↑ Precipitation risk: high Ca and phosphate concentrations**, **↑ pH** (low amino acid concentration), **↑ temperature**, **calcium chloride** (use gluconate), standing time, wrong mixing order, lipids hiding precipitate (TNA).
- **↓ Risk:** higher amino acid concentration (lower pH), adding **cysteine** (neonates), using **calcium gluconate**, **adding phosphate first and calcium near the end** with mixing, solubility curves, **in-line filters**.
- **Other additives: regular insulin** (only insulin type added; some adsorbs to the bag), famotidine (compatible), low-dose heparin • avoid adding other drugs (compatibility).

Worked PN calculation (example)

Step	Calculation (70 kg patient)
Total calories	25 kcal/kg × 70 = **1,750 kcal/day**
Protein	1.5 g/kg × 70 = **105 g** × 4 = 420 kcal
Lipid (~20% of calories)	350 kcal ÷ 2 kcal/mL = **175 mL of 20% lipid** (35 g)
Dextrose (remaining)	1,750 − 420 − 350 = 980 kcal ÷ 3.4 = **288 g**
Glucose infusion rate	288,000 mg ÷ 70 kg ÷ 1,440 min = **2.9 mg/kg/min** (✓ < 4–5)
Fluid	30–35 mL/kg = **2,100–2,450 mL/day**
Volumes of stock	Amino acid 10% (0.1 g/mL): 105 g → 1,050 mL • dextrose 70% (0.7 g/mL): 288 g → ~411 mL

Monitoring

Parameter	Frequency
Blood glucose	Every 6 h (or more) for the first 24–48 h, until stable
Electrolytes (Na, K, Mg, phosphate), BUN, SCr	**Daily** until stable (first 3 days, and closely if refeeding risk), then 1–2 times weekly
Triglycerides	Baseline and after starting lipids, then weekly
LFTs, CBC, INR	Baseline, then weekly
Weight, intake/output	Daily
Prealbumin, CRP, nitrogen balance	Weekly (interpret prealbumin with CRP)

Complications of PN

Type	Problems and management
Catheter-related	**Central line-associated bloodstream infection (CLABSI)** — most serious (S. epidermidis, S. aureus, **Candida**) • thrombosis • pneumothorax (insertion) • air embolism • **occlusion**: thrombus → **alteplase 2 mg** • lipid deposits → **70% ethanol** • Ca-phosphate precipitate → **0.1 N hydrochloric acid** • phlebitis (PPN)
Hyperglycemia	**Most common metabolic complication** → insulin in bag, ↓ dextrose, avoid overfeeding
Hypoglycemia	**Abrupt stop** (especially with insulin in the bag) → **taper** (e.g., halve rate for 1–2 h) or give D10W if PN stops suddenly
Electrolyte and acid–base disorders	Adjust additives daily • **refeeding syndrome** (see below)
Hypertriglyceridemia	TG > 400 → reduce/hold lipids • check propofol
Overfeeding	Hyperglycemia, **hepatic steatosis**, **hypercapnia**, azotemia (excess protein), fluid overload
PN-associated liver disease (PNALD / IFALD)	Adults: **steatosis** (dextrose overfeeding) • children: **cholestasis** (soybean phytosterols, no enteral stimulation) • prevention/treatment: **avoid overfeeding**, **cycle PN** (12–18 h), **limit soybean lipid ≤ 1 g/kg/day** or switch to **SMOF/fish oil**, start **enteral feeding**, **ursodiol**, reduce copper/manganese
Gallbladder	Sludge/stones (no enteral stimulation)
Gut atrophy	Loss of mucosal integrity → bacterial translocation (reason EN is preferred)
Metabolic bone disease	Long-term PN (calcium, vitamin D, aluminum)

- **Cycling PN** (home PN): infuse over 12–16 h • **taper up** the first hour and **taper down** the last hour to prevent hyper- and hypoglycemia.

6. Refeeding Syndrome

- Severe shifts in fluids and electrolytes when nutrition is restarted (oral, EN, or PN) in a malnourished patient → **insulin surge** drives **phosphate, potassium, and magnesium into cells** + ↑ thiamine use.
- Hallmark: **hypophosphatemia** • also **hypokalemia**, **hypomagnesemia**, **thiamine deficiency**, sodium and fluid retention.
- Usually within the **first 72 hours** (up to 5 days) • consequences: **arrhythmias**, **heart failure**, edema, **respiratory failure**, weakness, rhabdomyolysis, seizures, confusion, **death**.

Risk factors (ASPEN 2020)

- **BMI < 16** (high risk) or < 18.5 • **unintentional weight loss** > 15% in 3–6 months (or > 5–7.5% in 1–3 months) • **little or no intake for > 10 days** (or > 5–7 days with other risks) • **low K, phosphate, or Mg before feeding** • **alcohol use disorder**, **anorexia nervosa**, chronic malnutrition, cancer, post-bariatric surgery, uncontrolled diabetes, chronic diuretics/antacids.

Prevention and management

- **Start low**: **10–20 kcal/kg/day** (or ~100–150 g dextrose) and **advance by ~33% of goal every 1–2 days** over 3–7 days.
- **Check and correct K, phosphate, Mg before and during** feeding — monitor **every 12 h for the first 3 days** • correct while feeding (don't delay nutrition indefinitely for labs).
- **Thiamine 100 mg daily before** starting feeds and for **5–7 days** (or longer in high risk) • multivitamin daily.
- Monitor **vital signs, fluid balance, edema**, cardiac status • slow down nutrition if electrolytes fall sharply.

7. Special Populations

Population	Key points
Critically ill	Early EN (24–48 h) • trophic/hypocaloric first week (12–25 kcal/kg) • high protein (1.2–2 g/kg) • glucose 140–180 • count propofol calories
Obesity	**Hypocaloric, high-protein** (11–14 kcal/kg actual or 22–25 kcal/kg IBW; protein 2–2.5 g/kg IBW)
AKI/CKD	Don't restrict protein to avoid dialysis • ↑ protein on dialysis/CRRT • renal formulas (low K, phos) if electrolytes high • watch fluid
Liver disease	Protein 1.2–1.5 g/kg (no restriction in HE) • late-evening snack • ↓ copper/manganese in cholestasis
Acute pancreatitis	**Early oral/EN** (gastric or jejunal) preferred over PN
Burns, trauma	↑ calories and protein (up to 2–2.5 g/kg) • ↑ zinc, selenium, vitamin C
Short bowel syndrome	PN often needed • **teduglutide** (GLP-2 analog) ↓ PN dependence • PPIs, antidiarrheals, octreotide for high output
Diabetes	Glucose control (insulin in PN), diabetes-specific EN formula optional
Neonates/children	Higher calcium/phosphate needs (add **cysteine** to improve solubility) • pediatric AA products • lipid up to 3 g/kg/day • fish oil for cholestasis • aluminum limit < 5 mcg/kg/day

8. Nutrition-Related Drug Summary

Drug / product	Dose	Key ADRs	Key notes / interactions
Dextrose (PN)	Start 150–200 g/day; GIR ≤ 4–5 mg/kg/min	**Hyperglycemia**, steatosis, ↑ CO2	3.4 kcal/g
Amino acids (PN)	1.2–2 g/kg/day	Azotemia (excess)	4 kcal/g • specialty products rarely needed
Soybean lipid (Intralipid)	≤ 1–1.5 g/kg/day (20% = 2 kcal/mL)	**Hypertriglyceridemia**, PNALD, infection	CI severe egg/soy allergy • hold if TG > 400 • 12-h hang time alone
SMOFlipid / Clinolipid	≤ 2 g/kg/day	Hypertriglyceridemia	SMOF: fish allergy caution
Fish oil lipid (Omegaven)	1 g/kg/day (pediatric)	Bleeding (theoretical), EFAD if sole lipid long term	PN-associated cholestasis
Regular insulin (in PN)	~0.1 unit/g dextrose to start, titrate	**Hypoglycemia** (if PN stopped abruptly)	Only regular insulin in PN
Adult multivitamin (IV)	1 vial daily in PN	Rare (anaphylaxis — thiamine)	**Vitamin K 150 mcg** → ↓ warfarin effect
Trace elements	1 dose daily (adjust)	Toxicity: Mn (neuro), Cu (liver)	↓ Cu/Mn in cholestasis • ↓ Cr/Se in renal failure
Thiamine	100 mg daily (refeeding: before feeds, 5–7 days)	Rare	Give before glucose/feeds
Potassium / sodium phosphate	PN: 20–40 mmol/day • IV replacement 0.16–0.64 mmol/kg	Hyperkalemia (K-phos), hypocalcemia	**Ca-phos precipitation** • add phosphate first
Calcium gluconate (PN)	10–15 mEq/day	—	**Use gluconate** (not chloride) in PN • add near the end
Sodium/potassium acetate	As needed	Metabolic alkalosis (excess)	**Use acetate, never bicarbonate**, in PN
Metoclopramide	10 mg IV/PO q6h	**Tardive dyskinesia (boxed)**, EPS, QT	Max 12 weeks • renal dose adjustment

Drug / product	Dose	Key ADRs	Key notes / interactions
Erythromycin	250 mg IV q6–8h (≤ 72 h)	**QT**, GI cramps, tachyphylaxis	**CYP3A4 inhibitor**
Alteplase (Cathflo)	2 mg in catheter, dwell 30–120 min (repeat once)	Bleeding (minimal)	Thrombotic catheter occlusion
Ethanol 70% lock	Fill catheter volume	—	**Lipid** occlusion (also infection prevention)
Hydrochloric acid 0.1 N	Fill catheter volume	—	**Calcium-phosphate precipitate** occlusion
Ursodiol	10–15 mg/kg/day	Diarrhea	PN-associated cholestasis • separate from cholestyramine and antacids
Teduglutide	0.05 mg/kg SC daily	Abdominal pain, GI obstruction, ↑ polyp/neoplasia risk	Short bowel syndrome • colonoscopy before and during
Famotidine (in PN)	20 mg IV BID (renal adjust)	Confusion (elderly/renal)	Compatible additive
Pancreatic enzymes + sodium bicarbonate	1 crushed non-enteric tablet + 325 mg bicarbonate in water	—	**Unclogging feeding tubes**

Exam Traps and Frequently Tested Points

1. **"If the gut works, use it"** → EN preferred over PN • start EN within **24–48 h** in critically ill stable patients.
2. **Albumin and prealbumin** are negative acute-phase reactants → **not** reliable nutrition markers.
3. **Energy 25–30 kcal/kg/day** • ICU first week 12–25 • **protein 1.2–2 g/kg** (ICU) • 0.8 healthy • obese ICU: hypocaloric high-protein.
4. **Dextrose 3.4 kcal/g** • protein 4 • fat 9 • **lipid 20% = 2 kcal/mL** (10% = 1.1, 30% = 3) • **propofol 1.1 kcal/mL**.
5. **Nitrogen balance** = protein/6.25 − (UUN + 4).
6. **EN contraindications:** obstruction, severe ileus, high-output fistula, GI bleed, ischemia, shock.
7. **NG/NJ** short term (< 4–6 weeks) • **PEG/J-tube** long term • **post-pyloric = continuous only** • confirm placement by X-ray.
8. **Aspiration prevention:** head of bed **30–45°**, post-pyloric, prokinetics • don't routinely check GRV; don't hold for GRV < 500 mL.
9. **Diarrhea on EN → check sorbitol-containing meds, Mg, antibiotics, C. diff first**.
10. **Calorie-dense (1.5–2 kcal/mL)** for fluid restriction • renal formula: low K/phos • elemental for malabsorption • no pulmonary formula.
11. **Flush 30 mL water** before/after meds and every 4 h • unclog with warm water or pancreatic enzyme + bicarbonate (**not cola**).
12. **Don't crush EC, ER, SL, hazardous drugs** • give drugs separately • never in the feed bag.
13. **Phenytoin + tube feeds** → hold 1–2 h before/after • **warfarin** → INR changes • **quinolones** → chelation; no cipro suspension via tube • **levothyroxine** → separate.
14. **PN timing:** well-nourished low risk → wait 7 days • high risk/malnourished → start ASAP • supplemental PN after 7–10 days if EN < 60%.
15. **PPN ≤ 900 mOsm/L**, short term • CPN for full needs.
16. **Osmolarity:** dextrose 5 mOsm/g, amino acid 10 mOsm/g, electrolytes 2 mOsm/mEq.
17. **2-in-1 → 0.22-µm filter • TNA → 1.2-µm filter** • lipid alone hangs 12 h, TNA 24 h.
18. **Max GIR 4–5 mg/kg/min** • overfeeding dextrose → hyperglycemia, steatosis, ↑ CO2 (ventilator weaning).
19. **Regular insulin** is the only insulin added to PN • glucose goal 140–180.
20. **Hold lipids if TG > 400** • lipid CI: egg/soy (fish) allergy • EFAD: scaly rash, triene:tetraene > 0.2.
21. **Soybean lipid** (phytosterols) → PNALD/cholestasis → limit ≤ 1 g/kg/day, switch to SMOF/fish oil.

22. **Ca-phos precipitation** → fatal emboli • use **calcium gluconate**, add **phosphate first, calcium last** • ↑ risk with high pH, high temperature, low amino acids.
23. **Never add bicarbonate to PN** → use **acetate** for acidosis.
24. **Multivitamin has vitamin K** → warfarin • **↓ copper and manganese in cholestasis** • ↑ zinc with diarrhea/fistula.
25. **PN monitoring:** glucose q6h initially • electrolytes daily until stable • TG baseline and weekly • LFTs weekly.
26. **Catheter occlusion:** clot → **alteplase 2 mg** • lipid → **70% ethanol** • Ca-phos → **0.1 N HCl**.
27. **Most common PN metabolic complication: hyperglycemia** • most serious catheter complication: **CLABSI** (Candida).
28. **Abrupt PN stop** → hypoglycemia → taper.
29. **Refeeding syndrome: ↓ phosphate** (hallmark), ↓ K, ↓ Mg, thiamine deficiency • first 72 h • start **10–20 kcal/kg**, advance over 3–7 days, **thiamine 100 mg before feeding**, monitor electrolytes q12h × 3 days.
30. **Refeeding risk:** BMI < 16, weight loss > 15%, no intake > 10 days, low baseline electrolytes, alcoholism, anorexia.

> **Note:** Requirements and limits vary slightly between ASPEN guideline versions and institutions. **Use the numbers in the reference you are studying for the exam.**

Good luck on your exam!